

Project Initialization and Planning Phase

Date	30 September 2026
Team ID	SWTID-2026-8336
Project Title	EduGenie — Google Gemini Powered Learning Assistant
Maximum Marks	5 Marks

Project Proposal (Proposed Solution) report

The proposal report aims to transform loan approval using machine learning, boosting efficiency and accuracy. It tackles system inefficiencies, promising better operations, reduced risks, and happier customers. Key features include a machine learning-based credit model and real-time decision-making.

Project Overview	
Objective	The primary objective is to develop an AI-powered learning assistant that helps students understand academic topics, resolve doubts, practice through quizzes, and monitor their learning progress using Google Gemini.
Scope	The project provides a web-based learning platform with user authentication, AI-powered learning assistance, quiz generation and evaluation, learning history, study planning, and progress tracking.
Problem Statement	
Description	Students often face difficulty understanding academic topics, getting quick explanations for their doubts, finding suitable practice questions, and maintaining their learning progress. Existing resources may require students to use multiple platforms for different learning activities.
Impact	These difficulties can make learning less organized and reduce students' ability to practice and track their progress effectively. A unified learning assistant can provide accessible academic support and help students manage their learning activities in one platform.
Proposed Solution	
Approach	EduGenie uses Google Gemini to provide AI-powered academic assistance through a web application. The system connects the React.js frontend with a Node.js and Express.js backend, MongoDB Atlas for data storage, and the Google Gemini API for AI-powered learning support.

Key Features	• AI-powered academic assistance using Google Gemini • Student authentication and protected access • Learning history and study planning • Quiz generation and answer evaluation • Quiz results and progress tracking • Profile and settings management • ☐ Forgot-password support
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Benefits	- ☐ Provides quick and accessible AI-powered learning assistance. - ☐ Helps students understand topics through interactive learning support. - ☐ Enables practice through quizzes and answer evaluation. - ☐ Helps students review learning history and track progress. - ☐ Provides multiple learning functions through a single web application.
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Resource Requirements

Resource Type	Description	Specification/Allocation
Hardware		
Computing Resources	Laptop/Desktop computer suitable for web application development and testing.	T4 GPU
Memory	System memory required for running the frontend, backend, database tools and development environment.	8 GB
Storage	Local storage for source code, dependencies and project data.	1 TB SSD
Software		
Frameworks	React.js, Node.js, Express.js	Flask
Libraries	React Router, Axios, Mongoose, JWT, bcrypt	scikit-learn, pandas, numpy, matplotlib, seaborn
Development Environment	Visual Studio Code, Node.js, npm	Jupyter Notebook, pycharm
Data		
Data	MongoDB Atlas	Kaggle dataset, 614, csv UCI dataset, 690, csv